PROJECT PROPOSAL: PMJWAY – CAMPUS INTERACTIVE NAVIGATION & DIRECTORY SYSTEM

STUDENT & PROJECT INFORMATION

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- **Project Title:** PMJWay: Politeknik Mersing Campus Interactive Navigation & Directory System

1.0 Introduction & Project Background

Politeknik Mersing (PMJ) has a sprawling campus layout featuring multiple administration blocks, lecture halls, classrooms, as well as technical and computer laboratories. Following discussions with our project advisor, Madam Azrina, a fresh and innovative idea was sparked: to develop a mobile application called **PMJWay**.

The core inspiration behind this project stems from the need for an interactive digital directory system that mirrors modern navigation apps like Waze. Rather than relying on static maps, this system incorporates dynamic directional graphics (such as guiding arrows) and voice guidance features to help users seamlessly locate any room or laboratory across PMJ. This initiative is expected to significantly improve time management and navigation efficiency for both campus residents and visitors.

2.0 Problem Statement

Through initial observations and feedback, several key challenges at Politeknik Mersing were identified:

1. Difficulty in finding locations

New students face difficulty finding classes, rooms, and buildings due to an unfamiliarity with the PMJ campus layout.

2. Long search times

Students have to ask friends or staff for directions, resulting in wasted time and the risk of arriving late to class.

3. No centralized location guide

Information regarding the locations of classes and buildings is not provided in a single, easily accessible platform for new students.

3.0 Project Objectives

The objectives of this project are structured using SMART criteria (*Specific, Measurable, Achievable, Realistic, Time-oriented*):

1.Providing location information

Developing an application that provides location information for classes, rooms, buildings, and key facilities at PMJ.

2.Providing search and navigation functions

Enabling students to search for desired locations and obtain directions to those locations.

3.Assisting new students

Helping new students familiarize themselves with the campus environment and find locations more quickly and easily.

4.0 Project Scope

1. Campus Location Information

The application provides information on the locations of classes, rooms, buildings, offices, and key facilities available around PMJ.

2. Search and Navigation

Students can search for locations based on names or room numbers and obtain directions from their current location to the selected destination.

3. Target Users

The application focuses on new PMJ students, particularly those who are still unfamiliar with the layout and building locations on campus.

5.0 Project Significance & Social Responsibility

This project delivers substantial value to the Politeknik Mersing community:

- **Benefits for Students & Visitors:** Minimizes the time spent searching for classes or labs, thereby ensuring smoother academic activities.
- **Campus Service Innovation:** Empowers PMJ's digital ecosystem by providing a modern, accessible facility available to the entire campus community.
- **Addressing Lateness Issues:** Helps students overcome tardiness caused by difficulties in locating lecture rooms or technical lab spaces.

6.0 Project Literature Review

A literature review was conducted by evaluating existing systems and comparing their functional features systematically:

- **Comparative Analysis:** Traditional static maps are hard to read and entirely passive. While apps like Google Maps excel at outdoor road navigation, they lack indoor building details. PMJWay fills this gap by introducing a Waze-style interactive campus navigation feature.

Aspect	UUM Navigate: Campus Companion	UTHM U Finder	PMJ Campus Location App
OBJECTIVE	Assist new and lower-semester students in navigating the UUM campus and finding classrooms and campus facilities. (e-Journal)	Help new students and visitors navigate from their current location to destinations within UTHM. (AIP Publishing)	Help new PMJ students find classrooms, rooms, buildings and important facilities around the campus.
KEY FEATURES	Campus navigation, classroom location, campus facilities information and FAQ. (e-Journal)	Outdoor and indoor navigation, destination search, GPS, Google Maps and QR-code-based indoor positioning. (ResearchGate)	Location search, classroom and building information, campus map and directions to selected locations.
LOCATION SEARCH	Helps students locate classrooms and campus facilities. (e-Journal)	Users can select their desired destination from the application database. (ResearchGate)	Students can search for classrooms, rooms, buildings and selected campus facilities.
NAVIGATION	Provides campus navigation to assist students in finding locations around UUM. (e-Journal)	Provides both outdoor and indoor navigation. Outdoor navigation uses Google Maps, while indoor navigation uses QR codes and route images. (ResearchGate)	Provides simple navigation guidance to help students reach their selected classroom or building.
TARGET USERS	New students and lower-semester students at UUM. (e-Journal)	New students and visitors at UTHM. (AIP Publishing)	New students at PMJ, particularly students who are unfamiliar with the campus layout.
DEVELOPMENT / TECHNOLOGY	Developed using the **ADDIE model** consisting of Analysis, Design, Development,	Developed using **Thunkable**, with GPS/Google Maps for outdoor navigation and QR codes for indoor navigation. (ResearchGate)	Proposed mobile application developed specifically according to the location structure

	Implementation and Evaluation. (e-Journal)		and navigation needs of PMJ.
USER EVALUATION	Evaluated through expert review and a survey involving **35 UUM students**, focusing on usefulness, ease of use and satisfaction. (e-Journal)	The study focused on developing and demonstrating an indoor and outdoor campus navigation application for UTHM. (AIP Publishing)	Evaluation can focus on **ease of use, accuracy of location information and effectiveness in helping new students find their classrooms**.

7.0 Methodology

The project is developed using an **Agile / Prototyping Model**, breaking down the process into structured phases:

1. **Requirements Analysis Phase:** Collecting actual layout data (floor plans, room lists, and lab directories at PMJ) and gathering insights from our advisor.
2. **Design Phase:** Building user interface (UI/UX) wireframes, system flowcharts, and database schemas.
3. **Development Phase:** Coding the navigation system, integrating directional arrow graphics, and setting up the voice module.
4. **Testing Phase:** Conducting unit testing and integration testing to ensure application stability before final deployment.

8.0 Cost Planning

Estimated expenses and resources required for development are lean and efficient:

- **Development Software:** Utilizing open-source software and free coding tools.
- **Hardware & Testing:** Leveraging the team members' existing laptops and mobile devices.
- **Operational / Utility Costs:** Minimal costs covering internet access and test server maintenance.

9.0 Gantt Chart

Table1

No.	Fasa / Aktiviti Projek	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
1	Fasa Analisis & Perbincangan										
	• Perbincangan tajuk & skop Bersama Mdm Azrina	✓									
	• Pengumpulan data & senarai bilik/makmal PMJ	✓	✓								
2	Fasa Reka Bentuk (Design)										
	• Lukisan susun atur pelan lantai & ikon grafik		✓	✓							
	• Reka bentuk antara muka (UI/UX) & alur navigasi			✓	✓						
3	Fasa Pembangunan (Development)										
	• Pembangunan pangkalan data lokasi PMJ				✓	✓					
	• Pengekodan ciri carian & grafik anak panah					✓	✓				
	• Integrasi modul audio / panduan suara (Voice)						✓	✓			
4	Fasa Pengujian (Testing)										
	• Ujian unit & integrasi sistem (Unit/Integration)								✓		
	• Penambahbaikan ralat (Bug fixing & Usability)								✓	✓	
5	Fasa Dokumentasi & Penyerahan										
	• Penyediaan laporan teknikal akhir & slaid									✓	✓
	• Pembentangan projek (Project Presentation)										✓

10.0 References

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11.0 Conclusion

Overall, the proposed **PMJWay** project not only fulfills the academic requirements of the *Integrated Project* course but also solves real, everyday problems at Politeknik Mersing. By combining strong team leadership, close collaboration, innovative thinking, and social responsibility toward the campus community, this application is set to be successfully realized and bring long-term benefits to all PMJ citizens.